

# Athanasios Fytilis

🏠 29 March 2003 | 🏠 Giannitsa, Greece | 📄 Willing to Relocate

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## RESEARCH INTERESTS

Hardware and Software Architecture, Data Analysis and Structure, Systems Architecture, Operating Systems, Compilers, HPC and Embedded Systems.

## EDUCATION

Oct 2021 - Present Integrated Master's (Diploma) in COMPUTER SCIENCE AND ENGINEERING  
University of Ioannina, Ioannina, Greece

Oct 2018 - June 2021 Graduated High School with Excellence  
1st General Lyceum of Giannitsa, Giannitsa, Greece  
Commended thrice by the Hellenic Mathematical Society for distinguished performance in the "Thales-Euclid-Archimedes" mathematics competition (for each year of high school)

## WORK EXPERIENCE

Oct 2025-Present Full-Stack Developer at [FootyGreece](#), Remote  
*Full-stack engineering for a high-performance sports analytics and academy management platform*  
I develop scalable API endpoints, business logic layers, and data repositories using C# .NET 10 and Entity Framework Core. On the frontend, I engineer responsive views within a React 19 and TypeScript monorepo using Vite, Redux Toolkit, Redux Observable, and Zustand for optimized state management. I also implement real-time communication features by synchronizing SignalR hubs with frontend reactive programming (Epics), manage database schema updates through code-first migrations, and maintain cross-platform hybrid mobile app elements.

October 2022-2025 Member at [Student's Finance Club](#), University of Macedonia, Thessaloniki, Greece  
*Active member of the university's student-run financial intelligence and analytics club*  
I participated as a core contributor in a professional commodity analysis project on Crude Oil Futures in collaboration with VRS International S.A., conducting comprehensive technical indicator evaluations (CCI, MACD, Bollinger Bands) and fundamental market research. I also attended various conferences and speaking events concerning finance issues, while actively collaborating in student teams to learn how to analyze complex financial instruments and monitor global market trends effectively.

## PROJECTS

Spring semester 2026 JTAG & BIST SoC Testing Infrastructure, [GitHub Link](#)  
*VHDL, ModelSim, Quartus Prime*  
I designed a complete IEEE 1149.1 JTAG testing architecture, including a fully synchronous 16-state TAP Controller, Instruction Register, and Boundary Scan Cells (BSC) for internal logic isolation. I implemented an 8-bit Linear Feedback Shift Register (LFSR) for on-chip pseudo-random pattern generation and a 16-bit Galois MISR for signature analysis and output compaction. I also optimized the verification testbench using overlapping scan techniques, successfully reducing the required clock cycles per test vector from 9 to 5, and verified fault detection capabilities and Design for Testability (DFT) efficiency by identifying a stuck-at-1 fault through signature mismatch.

Spring Semester 2025 Greek++ Compiler (MY802), [GitHub Link](#)  
*Python 3, RISC-V Assembly, Compiler Design, AST, Systems Programming*  
I co-developed a complete, multi-stage compiler in Python 3 that translates a custom procedural language into executable RISC-V assembly. I implemented a lexical analyzer and a Recursive Descent parser to construct an Abstract Syntax Tree (AST) and perform rigorous semantic analysis using a custom Symbol Table. Furthermore, I engineered an intermediate code generator utilizing quadruples and backpatching for control flow optimization, and a final RISC-V code generator that successfully handles complex stack frame protocols and non-local variable access.

Spring Semester 2026 Food Hazard Detection (SemEval-2025), [GitHub Link](#)  
*Python, scikit-learn, Hugging Face (DistilBERT), NLP, Feature Engineering*  
I co-developed a hierarchical Natural Language Processing (NLP) system to classify food recall reports for the SemEval-2025 competition. To tackle extreme class imbalance, I engineered a robust LinearSVC classification pipeline utilizing FeatureUnion that significantly outperformed deep learning baselines (DistilBERT) in both accuracy and computational efficiency. Furthermore, I implemented custom feature engineering techniques, including Transductive TF-IDF, Character n-grams, and a novel Title Boosting mechanism. These optimizations effectively mitigated error propagation across hierarchical categories and achieved a highly competitive Macro-F1 ST1 score of 0.78003 on Kaggle, securing a 13.5% performance increase over the initial baseline.

## SKILLS

|                       |                                                                                                                                                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPOKEN LANGUAGES      | <b>English:</b> Professional Working Proficiency (LRN C2,C1), <b>Greek:</b> Native Speaker, <b>German:</b> A2 (still studying)                                                                                                                                                       |
| PROGRAMMING           | C/C++, C#, Python, Java 17+, TypeScript, JavaScript, Pthreads, OpenGL, MIPS Assembly, PostgreSQL, SQL Server, MySQL, H2 Database, MATLAB                                                                                                                                             |
| WEB/Frameworks        | Spring Boot, Spring Security, Spring Data JPA, .NET 10, ASP.NET Core, Entity Framework Core, React 19, Ionic, Capacitor, Zustand, Redux Toolkit, TailwindCSS, SignalR,HTML,CSS                                                                                                       |
| HARDWARE DESIGN TOOLS | Verilog/SystemVerilog, HLS C/C++, VHDL, Synopsys EDA Tools,Modelsim                                                                                                                                                                                                                  |
| SOFT SKILLS           | Prompt Engineering, GitHub, Git, Maven, Docker, Vite, Thymeleaf, Make, LaTeX (Overleaf/Prism), Visual Studio Code, Linux, Windows, Virtual Machines (VMware/VirtualBox),Insomnia ,Unity, AutoCAD, NI Multisim, Microsoft Office, Intel Advisor, Mininet, Electronic Circuits,Eclipse |

## ACADEMIC COURSES

### Completed

**MY803: Software Engineering** - Prof. Zarras Apostolos Grade 9/10  
Course Focus: Software Development Lifecycles, Requirements Analysis, Architectural Design & Quality Metrics

- Formulated architectural designs utilizing UML metrics and executed comprehensive white-box/black-box verification strategies across various SDLC models.

**MY406: Digital Design II** - Prof. Chrysovalantis Kavousianos Grade 8.5/10  
Course Focus: RTL Design, Synthesizable VHDL Modeling, Programmable Logic Devices (FPGAs/CPLDs)

- Engineered synthesizable digital circuits at the Register-Transfer Level (RTL) using VHDL, executing complete design workflows from behavioral/structural modeling to final placement and routing.

**D9: Natural Language Processing** - Prof. Constantine Skianis Grade 8.5/10  
Course Focus: Text Representation, Machine Learning, Neural Networks & Transformer Architectures

- Explored advanced NLP fundamentals and statistical models, utilizing modern neural networks and transformer architectures to deploy applications in sentiment analysis and machine translation.

**MY301: Software Development** - Prof. Panos Vassiliadis - Grade 8/10  
Course Focus: Object-Oriented Design, UML Modeling, Design Principles & Software Testing

- Developed large-scale software applying advanced Object-Oriented paradigms, SOLID principles, and rigorous Unit Testing methodologies via JUnit.

**MY404: Electronics** - Prof. George Tsiatouhas - Grade 7.5/10  
Course Focus: Circuit Theory, Semiconductor Physics, Diodes, BJT/FET Transistors & Multi-Stage Amplifiers

- Modeled the physical operation and I-V characteristics of BJT/FET transistors under DC biasing and evaluated fundamental multi-stage amplifier topologies.

**MYE041: Advanced Data Management** - Prof. Nikos Mamoulis - Grade 8/10  
Course Focus: Complex Data Types, Advanced Indexing Frameworks, R-Trees & Graph Query Processing

- Evaluated specialized indexing structures for spatial and high-dimensional data, and formulated execution strategies for advanced query paradigms, including top-k, skyline, and sub-graph matching algorithms.

**MYE010: Testing and Reliability of Electronic Systems** - Prof. Vassilios Tenentes Grade 7.5/10  
Course Focus: VLSI Testing, ATPG, Design for Testability (DFT), Scan Chains & BIST

- Analyzed VLSI testing methodologies, encompassing automatic test pattern generation (ATPG), fault simulation, and the architectural implementation of scan chains and Built-In Self-Test (BIST) structures.

**MY901: Microprocessors** - Prof. Vassilios Tenentes - Grade 7/10  
Course Focus: Low-Power Processor Design, Verilog RTL Simulation, Real-Time OS & Embedded C

- Simulated hardware modules at the register-transfer level (RTL) using Verilog and implemented low-level firmware for RTOS synchronization utilizing Assembly and Embedded C.